

AWS Manual Setup Guide

City Listing Project — ECS Fargate · RDS · ALB · Cloud Map

This guide walks you through manually deploying the City Listing stack on AWS using the console — no CloudFormation. Follow the sections in order; each one depends on the previous.

Layer	Service	Details
Internet entry	Application Load Balancer	Internet-facing, port 80, public subnets
Frontend	ECS Fargate — CityWeb	Port 8080, private subnets, attached to ALB
Backend API	ECS Fargate — CityApi	Port 8080, private subnets, Cloud Map discovery
Service discovery	Cloud Map	cityproject.local → api.cityproject.local:8080
Database	RDS SQL Server Express	db.t3.small, 20 GB gp2, private subnets
Observability	CloudWatch Logs	/ecs/CityProject-api + /ecs/CityProject-web

Deployment order

- 1 Security Groups
- 2 RDS SQL Server
- 3 CloudWatch Log Groups
- 4 Cloud Map namespace + service
- 5 Application Load Balancer + Target Group
- 6 IAM Role + ECS Cluster
- 7 ECS Task Definitions (API then Web)
- 8 ECS Services (API then Web)
- 9 Verify & test

Note: RDS must be running before you create the API task definition — you need the endpoint for the connection string. The ALB and target group must exist before creating the Web service.

Section 1 — Security Groups

Navigate to VPC → Security Groups → Create security group. Create all four groups below in your VPC before moving on.

ALB Security Group (city-alb-sg)

- **Inbound:** TCP 80 from 0.0.0.0/0
- **Outbound:** All traffic (default)

Web Security Group (city-web-sg)

- **Inbound:** TCP 8080 from city-alb-sg
- **Outbound:** All traffic

API Security Group (city-api-sg)

- **Inbound:** TCP 8080 from city-web-sg
- **Outbound:** All traffic

RDS Security Group (city-rds-sg)

- **Inbound:** TCP 1433 from city-api-sg
- **Outbound:** All traffic

Note: Always use security group references as source (not CIDR) for service-to-service rules — this is the chain that locks down lateral movement.

Section 2 — RDS SQL Server

Navigate to RDS → Create database. This takes 10–15 minutes; start it early and move on to the next sections while it provisions.

Parameter	Value / Action
Engine	Microsoft SQL Server → Express edition
Template	Free tier or Dev/Test
DB instance ID	city-database
Master username	admin
Master password	Min 8 chars — must include uppercase, lowercase, digit, special char
Instance class	db.t3.small
Storage	20 GB, gp2, no autoscaling required
Multi-AZ	No
VPC	Your VPC
Subnet group	Create new — select your two private subnets
Public access	No

VPC security group	city-rds-sg
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Note: Copy the endpoint address once the instance shows 'Available' — you'll need it for the API connection string in Section 7.

Section 3 — CloudWatch Log Groups

Navigate to CloudWatch → Log groups → Create log group. Create both groups:

Parameter	Value / Action
/ecs/CityProject-api	Retention: 7 days
/ecs/CityProject-web	Retention: 7 days

Section 4 — Cloud Map — Service Discovery

Navigate to Cloud Map → Create namespace.

Namespace

Parameter	Value / Action
Namespace name	cityproject.local
Type	API calls and DNS queries in VPCs
VPC	Your VPC

Service (inside the namespace)

Go into the new namespace → Create service:

Parameter	Value / Action
Service name	api
DNS routing policy	Multivalue
Record type	A, TTL: 60
Health check	Custom (ECS manages this), failure threshold: 1

Note: This creates the DNS name api.cityproject.local which the CityWeb container uses to reach CityApi.

Section 5 — Application Load Balancer + Target Group

Navigate to EC2 → Load Balancers → Create load balancer → Application Load Balancer.

Target Group — create this first

Click 'Create target group' (opens in a new tab). Do not register any targets yet — ECS will do this automatically.

Parameter	Value / Action
Target type	IP addresses
Name	CityProject-WebTG
Protocol / Port	HTTP / 8080
VPC	Your VPC
Health check path	/
Interval	30 s
Timeout	5 s
Healthy threshold	2
Unhealthy threshold	3

Application Load Balancer

Parameter	Value / Action
Name	CityProject-ALB
Scheme	Internet-facing
IP type	IPv4
VPC	Your VPC
Subnets	Both public subnets
Security group	city-alb-sg
Listener	HTTP / 80 → forward to CityProject-WebTG

Section 6 — IAM Role + ECS Cluster

IAM Execution Role

Navigate to IAM → Roles → Create role:

Parameter	Value / Action
Trusted entity	AWS service → Elastic Container Service → ECS Task
Attached policy	AmazonECSTaskExecutionRolePolicy

Role name	CityProject-EcsTaskExecutionRole
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ECS Cluster

Navigate to ECS → Clusters → Create cluster:

Parameter	Value / Action
Cluster name	CityProjectCluster
Infrastructure	AWS Fargate (serverless)

Section 7 — ECS Task Definitions

Navigate to ECS → Task definitions → Create new task definition. Create API first, then Web.

API Task Definition

Parameter	Value / Action
Family name	CityProject-ApiTask
Launch type	AWS Fargate
CPU / Memory	0.25 vCPU / 0.5 GB
Execution role	CityProject-EcsTaskExecutionRole
Container name	city-api
Image URI	Your ECR URI for the API image
Port mapping	8080 / TCP
Env var name	ConnectionStrings__DefaultConnection
Env var value	Server=<RDS_ENDPOINT>,1433;Database=CityDb;UserId=admin;Password=<PASSWORD>;TrustServerCertificate=True
Log driver	awslogs
awslogs-group	/ecs/CityProject-api
awslogs-region	Your AWS region
awslogs-stream-prefix	ecs

Web Task Definition

Parameter	Value / Action
Family name	CityProject-WebTask
Launch type	AWS Fargate
CPU / Memory	0.25 vCPU / 0.5 GB
Execution role	CityProject-EcsTaskExecutionRole

Container name	city-web
Image URI	Your ECR URI for the Web image
Port mapping	8080 / TCP
Env var name	ApiSettings__CityApiBaseUrl
Env var value	http://api.cityproject.local:8080
Log driver	awslogs
awslogs-group	/ecs/CityProject-web
awslogs-region	Your AWS region
awslogs-stream-prefix	ecs

Section 8 — ECS Services

Navigate to ECS → CityProjectCluster → Services → Create. Deploy API first, then Web.

API Service

Parameter	Value / Action
Launch type	FARGATE
Task definition	CityProject-ApiTask (latest)
Service name	CityProject-ApiService
Desired tasks	1
VPC	Your VPC
Subnets	Both private subnets
Security group	city-api-sg
Public IP	ENABLED if no NAT Gateway · DISABLED if NAT Gateway exists
Service discovery	Turn ON → namespace: cityproject.local → existing service: api
Load balancer	None

Web Service

Parameter	Value / Action
Launch type	FARGATE
Task definition	CityProject-WebTask (latest)
Service name	CityProject-WebService
Desired tasks	1
VPC	Your VPC
Subnets	Both private subnets
Security group	city-web-sg
Public IP	Same choice as API Service above
Load balancer	Application Load Balancer → CityProject-ALB
Listener	Port 80 (existing)
Target group	CityProject-WebTG (existing)
Service discovery	OFF — not needed for the web tier

Section 9 — Verify & Test

Follow these steps to confirm the stack is healthy end-to-end:

- Go to ECS → CityProjectCluster → Services — both services should show tasks in RUNNING state.
- Go to EC2 → Load Balancers → copy the DNS name of CityProject-ALB.
- Open http://<ALB_DNS_NAME> in your browser — you should see the CityWeb frontend.
- Go to CloudWatch → Log groups → open /ecs/CityProject-api and /ecs/CityProject-web to confirm logs are flowing.

Troubleshooting

Tasks keep stopping

ECS → cluster → service → Tasks tab → click a stopped task → expand the container to read the stop reason.

Common causes: wrong ECR image URI, bad connection string, or RDS not yet available.

CREATE_FAILED on RDS

Password doesn't meet SQL Server complexity requirements. Must include uppercase, lowercase, digit, and special character.

Web can't reach API

Verify the security group chain: city-alb-sg → city-web-sg → city-api-sg. Also confirm Cloud Map service is registered and the API task is healthy.

Tasks can't pull ECR images

If private subnets have no NAT Gateway, set AssignPublicIp = ENABLED and use public subnets, OR add a VPC endpoint for ECR (com.amazonaws..ecr.dkr).

Target group shows unhealthy

Check the health check path '/' returns HTTP 200 from CityWeb. Also verify the Web security group allows inbound 8080 from city-alb-sg.